

Lecture 5 Notes:

Logicism: Is Mathematics (Just) Logic?

Philosophy of Mathematics

Central Theme

Logicism is the view that mathematics, or at least a significant part of mathematics, can be reduced to logic.

Mathematics = logic + definitions?

The guiding question:

Can mathematics be analytic rather than synthetic?

Logicism offers a third route after Kant and Mill.

Kant	Mill	Logicism
synthetic a priori intuition	synthetic a posteriori experience	analytic a priori logic and definitions

1. The Logicist Alternative

Kant held that mathematics is synthetic a priori. Mathematical knowledge is not derived from experience, but it is also not merely conceptual analysis.

Mill took the opposite route. Mathematics is synthetic, but empirical. It is grounded in experience and induction.

Logicism proposes another answer:

Mathematics is analytic.

Mathematical truths are knowable a priori because they can be derived from logic and definitions alone.

Teaching Point

Logicism tries to preserve the necessity and a priority of mathematics without Kantian intuition and without Millian empiricism.

Whiteboard

Kant: intuition

Mill: experience

Logicism: logic + definitions

2. Frege's New Conception of Analyticity

For Kant, a proposition is analytic if the predicate concept is contained in the subject concept.

Frege changes the issue. For Frege, analyticity concerns the ultimate ground of justification.

A truth is analytic if it is provable from logic and definitions alone.

So the question is not whether $7 + 5 = 12$ is contained in the concepts of 7, 5, +, and 12. The question is whether arithmetic can be derived from logical laws and definitions.

Teaching Point

Frege turns analyticity from a theory of concept-containment into a theory of proof.

Whiteboard

Kant: analytic = concept containment

Frege: analytic = proof from logic and definitions

3. Frege's Program

Frege's project is:

Define number logically, then derive arithmetic logically.

This makes arithmetic a test case for logicism.

Frege is a realist about mathematics. He thinks numbers are genuine objects and that mathematical statements have objective truth-values.

Realism in ontology	Numbers exist as objects.
Realism in truth-value	Arithmetic statements are objectively true or false.

Teaching Point

Frege is not trying to eliminate numbers. He is trying to show that numbers are logical objects.

Whiteboard

Number $\rightarrow$ logical object

Arithmetic $\rightarrow$ logical science

4. Equinumerosity

Frege begins with a basic fact about counting.

Two concepts are equinumerous if the objects falling under one can be put into one-to-one correspondence with the objects falling under the other.

Examples:

napkins $\sim$ plates

if there is exactly one napkin for each plate.

husbands $\sim$ wives

in a monogamous society, by definition.

Teaching Point

Counting can be understood through one-to-one correspondence before we explicitly introduce numbers.

Whiteboard

$F \sim G$ iff F s and G s can be paired one-to-one.

same number = equinumerous

5. Hume's Principle

Frege's central principle is now called Hume's Principle.

$$\boxed{\#F = \#G \text{ iff } F \sim G}$$

In words:

The number of F = the number of G

if and only if

F and G are equinumerous.

This principle connects number-identity with one-to-one correspondence.

Teaching Point

Hume's Principle is the heart of Fregean logicism. It gives identity conditions for numbers in terms of equinumerosity.

Whiteboard

$$\#F = \#G \iff F \sim G$$

number identity $\iff$ one-to-one correspondence

6. Constructing the Natural Numbers

Frege defines zero using an impossible concept.

$$Zx \equiv x \neq x$$

Since nothing is non-self-identical, Z applies to no objects.

$$0 = \#Z$$

Then one is the number of the concept "identical with zero":

$$1 = \#(x = 0)$$

Two is the number of the concept "identical with zero or identical with one":

$$2 = \#(x = 0 \vee x = 1)$$

And so on.

Teaching Point

Frege tries to generate the natural numbers from logical notions: object, concept, identity, and equinumerosity.

Whiteboard

$$0 = \#(x \neq x)$$

$$1 = \#(x = 0)$$

$$2 = \#(x = 0 \vee x = 1)$$

$$3 = \#(x = 0 \vee x = 1 \vee x = 2)$$

7. Successor and Natural Number

Frege defines the successor relation without presupposing arithmetic.

Roughly:

n is the successor of m

if there is a concept F such that F applies to exactly n objects, and removing one object from F leaves exactly m objects.

$$n = m + 1$$

if n is the number of a concept obtained by adding one object to a concept with number m .
Frege then defines natural number without circularly appealing to “finitely many.”

n is natural iff n belongs to every concept containing 0 and closed under successor.

Teaching Point

Frege’s definition of natural number anticipates the structure of mathematical induction.

Whiteboard

$$0 \in F$$

$$\forall n(n \in F \rightarrow S(n) \in F)$$

$\therefore$ every natural number belongs to F

8. Frege’s Theorem

Frege derives the basic laws of arithmetic from Hume’s Principle and second-order logic.

This result is now called Frege’s Theorem.

$\text{Hume's Principle} + \text{second-order logic} \Rightarrow \text{arithmetic}$

The theorem is a major mathematical and philosophical achievement.

Teaching Point

Even if Frege’s full system fails, Frege’s Theorem shows that arithmetic has a surprisingly deep connection with logic and abstraction.

Whiteboard

Hume's Principle

+

Second-order logic

⇓

Arithmetic

9. The Caesar Problem

Frege was not satisfied with Hume's Principle alone.

Hume's Principle settles identities of this form:

$$\#F = \#G$$

But it does not settle identities of this form:

$$\#F = t$$

where t is an arbitrary singular term.

For example:

$$\#F = \text{Julius Caesar?}$$

This is the Caesar problem.

The question is not whether anyone will actually confuse the number 2 with Julius Caesar. The question is whether Frege has given full identity conditions for numbers as objects.

Teaching Point

If numbers are objects, we need to know when a number is identical with, or distinct from, any object whatsoever.

Whiteboard

$$\#F = \#G \quad \text{settled}$$

$$\#F = \text{Caesar} \quad \text{not settled}$$

No entity without identity.

10. Extensions and Basic Law V

Frege tries to solve the Caesar problem by defining numbers as extensions of concepts.

$\#F$ = the extension of the concept equinumerous with F

So the number two is, roughly, the extension containing all concepts that apply to exactly two objects.

Frege's system depends on Basic Law V.

$$\boxed{\text{Ext}(F) = \text{Ext}(G) \quad \text{iff} \quad \forall x(Fx \leftrightarrow Gx)}$$

In words: two concepts have the same extension if and only if they apply to exactly the same objects.

Teaching Point

Frege's logicism depends on a theory of extensions. That theory is where disaster strikes.

Whiteboard

$$\text{Ext}(F) = \text{Ext}(G) \iff \forall x(Fx \leftrightarrow Gx)$$

11. Russell's Paradox

Russell discovered that Basic Law V is inconsistent.

A simplified version of the paradox uses the class:

$$R = \{x : x \notin x\}$$

Now ask:

$$R \in R?$$

If $R \in R$, then by definition $R \notin R$.

If $R \notin R$, then by definition $R \in R$.

So:

$$R \in R \iff R \notin R$$

Contradiction.

Teaching Point

Russell's paradox destroyed Frege's original logicist system, because it showed that Frege's theory of extensions was inconsistent.

Whiteboard

$$R = \{x : x \notin x\}$$

$$R \in R?$$

$$\text{Yes} \Rightarrow \text{No}$$

$$\text{No} \Rightarrow \text{Yes}$$

12. Russell's Response: The Vicious Circle Principle

Russell diagnoses the paradox as the result of illegitimate circularity.

No totality may contain members defined in terms of that very totality.

This is the vicious circle principle.

It rules out impredicative definitions: definitions that define something by referring to a totality that already contains it.

Teaching Point

Russell's response to paradox is to restrict what can meaningfully be defined.

Whiteboard

Impredicative definition

=

definition using a totality containing the thing defined

Russell: ban it.

13. Type Theory

Russell introduces type theory to block self-membership and vicious circularity.

Type 0	individuals
Type 1	classes of individuals
Type 2	classes of classes of individuals
Type 3	classes of type 2 classes

A class cannot be a member of itself because it is always of a higher type than its members.

$$x \in C$$

is meaningful only when C is one type higher than x .

Teaching Point

Type theory avoids Russell's paradox by making self-membership meaningless.

Whiteboard

individuals $\rightarrow$ classes $\rightarrow$ classes of classes $\rightarrow$ classes of classes of classes

$R \in R$ is not false; it is ill-typed.

14. Russell's Numbers

Russell defines numbers as classes of equinumerous classes.

2 = the class of all pairs

3 = the class of all triples

So the number of a class C is the class of all classes equinumerous with C .

This resembles Frege, but Russell's type restrictions create complications.

Teaching Point

Russell preserves the general Fregean idea that numbers are tied to equinumerosity, but he changes the logical framework to avoid paradox.

Whiteboard

0 = class of all empty classes

1 = class of all singletons

2 = class of all pairs

15. The Axiom of Infinity

Russell cannot prove that there are infinitely many natural numbers.

If there are only finitely many individuals, then there are only finitely many classes of individuals, and the natural numbers run out.

So Russell and Whitehead introduce the axiom of infinity.

There are infinitely many individuals.

But this axiom does not look like a logical truth.

Teaching Point

Russell avoids Frege's paradox, but he must add assumptions that weaken the logicist claim.

Whiteboard

Frege: proves infinity, but paradox.

Russell: avoids paradox, but assumes infinity.

16. The Axiom of Reducibility

Ramified type theory also blocks many definitions needed for ordinary mathematics.

To recover mathematics, Russell and Whitehead introduce the axiom of reducibility.

Roughly:

For every class, there is an equivalent predicative class.

This allows them to proceed as if many impredicative definitions were acceptable.

But the axiom looks ad hoc. It is introduced because mathematics needs it, not because it is obviously logical.

Teaching Point

The axiom of reducibility reveals the cost of Russell's logicism: to save mathematics, he must add principles that do not seem purely logical.

Whiteboard

Avoid paradox $\rightarrow$ type restrictions

Recover mathematics $\rightarrow$ reducibility

Question: is this still logicism?

17. Russell's No-Class Theory

Russell later denies that classes are genuine objects.

Classes are logical fictions.

Since numbers are constructed from classes, numbers also become logical fictions.

Frege	Russell
numbers are logical objects	numbers are logical fictions
realist ontology	anti-realist about classes
extensions as objects	classes eliminated by paraphrase

Teaching Point

Frege and Russell are both logicians, but they differ sharply about what numbers are.

Whiteboard

Frege: numbers exist as logical objects.

Russell: numbers are logical fictions.

18. Carnap and Logical Positivism

Carnap shifts the debate from ontology to language.

He thinks the traditional question:

Do numbers really exist?

is either trivial or meaningless, depending on how it is asked.

Carnap distinguishes internal and external questions.

Internal questions	External questions
asked within a framework	asked about the framework as a whole
answered by rules	pragmatic or pseudo-question
Is there a prime greater than 100?	Do numbers really exist?

Teaching Point

For Carnap, mathematics is analytic within a linguistic framework. The metaphysical question of whether numbers really exist is not a genuine factual question.

Whiteboard

Internal: $\exists n(n \text{ is prime and } n > 100)$

External: Do numbers really exist?

Carnap: choose frameworks pragmatically.

19. Carnap's Principle of Tolerance

Carnap defends a principle of tolerance.

We are free to adopt whatever linguistic framework is useful, provided the rules are clear.

The question is not:

Is this framework metaphysically correct?

The question is:

Is this framework useful for science and inquiry?

This lets Carnap accept impredicative systems if they are useful, without defending them by metaphysics.

Teaching Point

Carnap replaces metaphysical justification with pragmatic choice of language.

Whiteboard

Frameworks are tools.

Choose by clarity, efficiency, and usefulness.

No metaphysical tribunal.

20. Ayer and Mathematical Truth

Ayer states the positivist view clearly.

Empiricism says that factual truths are known through experience.

But mathematics seems necessary and certain.

So the positivist concludes:

Mathematics is necessary because it has no factual content.

Mathematical truths are analytic. They are true by virtue of linguistic rules.

Science	Mathematics
factual	analytic
empirically testable	true by rules
fallible	necessary

Teaching Point

Logical positivism preserves empiricism by saying that mathematics is not factual.

Whiteboard

If factual, then empirical.

Mathematics is not empirical.

$\therefore$ mathematics is not factual.

21. Geometry as a Framework

Logical positivists also treat geometry as a framework.

Euclidean geometry is not refuted by measuring a triangle and finding that its angles do not sum to 180° .

Instead, we say:

the measurement was wrong

or

the physical triangle is not Euclidean.

Pure geometry is analytic within its framework.

The empirical question is which geometry is useful for physics.

Teaching Point

The logical positivists separate pure mathematics from applied mathematics.

Whiteboard

Pure geometry: analytic framework

Applied geometry: empirical choice

Euclidean? Non-Euclidean?

22. Problems for Logical Positivism

Logical positivism eventually runs into several major problems.

Problem	Question
Verification	What exactly makes a factual statement verifiable?
Self-reference	Is the verification principle itself analytic or empirically verifiable?
Analytic/synthetic distinction	Can we sharply distinguish truths by meaning from truths by fact?
Mathematical practice	Is mathematics really divided into self-contained frameworks?

Quine attacks the analytic/synthetic distinction and develops a more holistic empiricism.

Teaching Point

The positivist account of mathematics depends on a sharp distinction between analytic framework rules and empirical factual claims. That distinction comes under pressure.

Whiteboard

Analytic | Synthetic

Quine: no sharp boundary.

23. Gödel and Self-Contained Frameworks

Carnap's picture suggests that mathematical truths are knowable from the rules of their framework.

But Gödel's incompleteness theorem complicates this.

If D is a sufficiently strong effective deductive system, then there are sentences in the language of D that are not decided by the rules of D .

$$D \not\vdash P \quad D \not\vdash \neg P$$

In practice, mathematicians often prove results about one structure by embedding it in a richer structure.

Teaching Point

Mathematical knowledge is not neatly confined inside isolated linguistic frameworks.

Whiteboard

Arithmetic statement

↓

proved using richer mathematics

Frameworks are not self-contained.

24. Fermat's Last Theorem

Fermat's Last Theorem is a useful example.

It is a statement about natural numbers:

$$a^n + b^n \neq c^n$$

for positive natural numbers a, b, c and $n > 2$.

The statement is elementary enough to understand with basic arithmetic.

But its proof uses highly sophisticated mathematics far beyond elementary arithmetic.

Teaching Point

Understanding the statement of a theorem is not the same as having the resources to know that it is true.

Whiteboard

Simple statement

$$a^n + b^n \neq c^n$$

Deep proof

elliptic curves, modular forms, advanced structures

25. Neo-Logicism

Contemporary neo-logicians, such as Crispin Wright and Bob Hale, revive Frege's project.

They avoid Basic Law V and work directly with Hume's Principle.

$$\boxed{\#F = \#G \text{ iff } F \sim G}$$

The neo-logician claim is not usually that Hume's Principle is a logical truth in the strict old sense.

Rather, Hume's Principle is treated as analytic, or meaning-constitutive, of the concept of cardinal number.

Hume's Principle + second-order logic $\Rightarrow$ arithmetic

Teaching Point

Neo-logicism preserves Frege's central insight while avoiding the inconsistent theory of extensions.

Whiteboard

Frege: Basic Law V $\Rightarrow$ paradox

Neo-logicism: Hume's Principle $\Rightarrow$ arithmetic

26. Problems for Neo-Logicism

Neo-logicism faces several major problems.

Problem	Question
Second-order logic	Is second-order logic really logic, or is it set theory in disguise?
Caesar problem	Does Hume's Principle fully identify numbers as objects?
Bad company	How do we distinguish good abstraction principles from bad ones?
Scope	Can the method extend beyond arithmetic?

The bad company objection is especially important.

Basic Law V and Hume's Principle are both abstraction principles. One is inconsistent; the other is fruitful.

What separates good abstraction from bad abstraction?

Teaching Point

Neo-logicism depends on showing that Hume's Principle is legitimate in a way that bad abstraction principles are not.

Whiteboard

Good abstraction: Hume's Principle

Bad abstraction: Basic Law V

Need a non-ad hoc distinction.

Conclusion: What Would Logicism Have to Show?

Logicism must show that mathematics can be derived from logic and definitions.

But the chapter repeatedly presses the same question:

What counts as logic?

If logic includes concepts, extensions, classes, second-order quantification, infinity, choice, reducibility, and abstraction principles, then the reduction may look suspicious.

Perhaps mathematics has not been reduced to logic.

Perhaps mathematics has been moved into logic.

Frege	Russell	Carnap
numbers as logical objects	numbers as logical fictions	numbers as framework-internal entities
realist logicism	type-theoretic logicism	linguistic logicism
paradox problem	extra axioms problem	framework problem

Final Framing

To teach this chapter well, present logicism as a powerful but unstable answer to the post-Kantian problem.

It gives a beautiful explanation of the a priori character of mathematics:

Mathematics is knowable because it follows from logic.

But it also faces a deep challenge:

The more logic must contain, the less clear it is that mathematics has been reduced.

Logicism lives or dies by the boundary between logic and mathematics.

Discussion Questions

1. What does logicism claim about mathematics?
2. How does Frege's notion of analyticity differ from Kant's?
3. Why is equinumerosity important for defining number?
4. What does Hume's Principle say?
5. Does Hume's Principle fully identify numbers as objects?
6. What is the Caesar problem?
7. Why does Russell's paradox undermine Frege's system?
8. Does type theory solve the paradox at too high a cost?
9. Is the axiom of infinity a logical truth?
10. Why does the axiom of reducibility look ad hoc?
11. Are numbers objects, fictions, or framework-internal entities?
12. Is Carnap right that external ontological questions are pseudo-questions?
13. Does Gödel's incompleteness theorem undermine the idea of self-contained mathematical frameworks?
14. Is second-order logic really logic?
15. Can neo-logicism succeed where Frege's original logicism failed?

Key Terms

Logicism The view that mathematics, or a significant part of mathematics, can be reduced to logic.

Analytic truth For Frege, a truth provable from logic and definitions alone.

A priori Knowable independently of sensory experience.

Concept For Frege, something under which objects fall; roughly, a property.

Equinumerosity The relation between two concepts or collections when their objects can be placed in one-to-one correspondence.

Hume's Principle The principle that the number of F is identical with the number of G if and only if F and G are equinumerous.

Frege's Theorem The result that arithmetic can be derived from Hume's Principle in second-order logic.

Caesar problem The problem that Hume's Principle determines identities between numbers but not identities between numbers and arbitrary objects.

Extension The class of all objects falling under a concept.

Basic Law V Frege's principle that two concepts have the same extension if and only if they apply to the same objects.

Russell's paradox The contradiction generated by considering the class of all classes that are not members of themselves.

Impredicative definition A definition that defines an object by referring to a totality containing that object.

Vicious circle principle Russell's principle that no object may be defined by reference to a totality containing that object.

Type theory Russell's hierarchy of individuals, classes of individuals, classes of classes, and so on, designed to avoid paradox.

Axiom of infinity Russell and Whitehead's postulate that there are infinitely many individuals.

Axiom of reducibility Russell and Whitehead's postulate that every class is equivalent to a predicative class.

Logical fiction Russell's term for entities, such as classes, that are useful in discourse but not part of the ultimate furniture of reality.

Linguistic framework Carnap's term for a system of language and rules within which questions can be meaningfully asked and answered.

Internal question For Carnap, a question asked within a linguistic framework.

External question For Carnap, a question about the reality or legitimacy of a framework as a whole.

Principle of tolerance Carnap's view that we may choose whichever linguistic frameworks are useful, provided their rules are clear.

Neo-logicism A contemporary revival of Fregean logicism, especially through Hume's Principle and abstraction principles.

Bad company objection The objection that neo-logicians need a principled way to distinguish acceptable abstraction principles from unacceptable ones.